

SMS Identity Verification Service (SMS-IVS)

Version 1.0 - Core Security Service

Overview

Centralized backend service for SMS OTP verification. All protected modules use this shared service.

Objectives

Generate, send, validate and audit SMS OTPs with centralized policies.

Business Rules

6-digit OTP; 5-minute expiry; 60-second resend cooldown; max 5 attempts; configurable daily limits.

Verification Triggers

Registration, password reset/change, phone number change, new device login, suspicious login, account recovery and sensitive account actions.

Lifecycle

Request → Generate OTP → Hash → Send SMS → Verify → Validate → Mark verified → Complete requested action.

Database

OTP Requests, Trusted Devices and SMS Audit Log tables.

API

POST /api/v1/sms/request; POST /api/v1/sms/verify; POST /api/v1/sms/resend; GET /api/v1/sms/status.

Administration

Configure OTP policies, SMS provider, sender ID and monitor delivery metrics. Never expose OTP values.

Security

Hash OTPs, single-use, expiration, rate limiting, brute-force protection, audit logging.

Backend Structure

src/sms-verification/{controllers,services,providers,repositories,dto,entities,policies,validators,event
s,monitoring}/sms-verification.module.ts

Recommendation

Use SMS-IVS as a reusable shared service across the entire backend.